RED SEAL EXAM PREP

442A Industrial Electrician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings

https://redsealquiz.ca · Edition 2026-07

About This Material

Independent, unofficial study resource. This publication is not affiliated with, endorsed by, or approved by the Canadian Council of Directors of Apprenticeship (CCDA), Employment and Social Development Canada, the Red Seal Program, Skilled Trades Ontario, or any provincial or territorial apprenticeship authority. “Red Seal” is used solely to describe the examination for which this material provides practice.

Originality. All questions and explanations in this publication are original works created by the author. This material references only publicly available occupational standards (Red Seal Occupational Standard) for topic alignment and reproduces no content from any actual examination.

No guarantee of results. This study material is provided for educational and informational purposes only. Success on any certification examination depends on many factors, and no representation is made that using this material will result in passing any exam. Exam content, structure, and weightings are set by certification authorities and may change without notice.

Technical accuracy. Code values and technical specifications cited in explanations reflect commonly referenced Canadian standards at the time of writing. Always defer to the current edition of the applicable code or the manufacturer's documentation in real work.

Refunds. Because this is an instantly delivered digital file, all sales are final once the download link has been delivered. If you hit a technical problem — a corrupted or inaccessible file, or a duplicate purchase — contact us within 14 days and we will repair the file or refund you in full.

Updates. Corrections and additions are published to the same file. Re-download from your Ko-fi library to get the current edition at no extra cost.

Mock Exam — answer all questions, then check the key

Q1. What does it mean to force a PLC input or output?

A) It raises the module's output rating so a heavier load can be driven
B) It stores the running program in the processor's non-volatile memory
C) It holds the processor in run mode until the force is removed
D) It overrides the point's real state, so logic and field disagree

Q2. In an AC circuit, what is power factor?

A) The ratio of apparent power to real power
B) The ratio of reactive power to apparent power
C) The ratio of real power to apparent power
D) The ratio of real power to reactive power

Q3. A 442A electrician finds a 3-phase induction motor drawing Phase A=28A, Phase B=31A, Phase C=24A while running unloaded at rated voltage. What is the MOST likely cause?

A) Voltage unbalance at the supply terminals
B) Worn motor bearings causing mechanical drag
C) Single-phasing from a blown control fuse
D) Motor is overloaded beyond nameplate rating

Q4. What is a current transformer (CT) and how is it used in metering?

A) A transformer reducing primary current to 5A or 1A for metering
B) A transformer used for variable current control in motor drives
C) A transformer that converts AC to DC for measurement
D) A clamp meter that measures current without contact

Q5. Before touching conductors that have been isolated and locked out, a worker tests for absence of voltage. Why must the tester be proved on a known live source both before and after that test?

A) To prove that the tester itself was working when it read zero
B) To confirm the tester is set to the correct voltage range first
C) To satisfy the requirement that two workers witness the test
D) To discharge any stored charge left in the tester's own leads

Q6. A technician performs a voltage drop test on a 120V, 20A branch circuit supplying a load 50m from the panel. The voltage drop measured is 8V (6.7%). What corrective action should be taken per CEC requirements?

A) Add a voltage regulator at the load end
B) Replace the load with a lower-power model
C) No action needed — CEC allows 10% drop
D) Increase the conductor size for the run

Q7. **A 442A electrician performs a polarization index test on a motor stator winding. The insulation resistance reads 80 MΩ at one minute and 120 MΩ at ten minutes. What is the polarization index, and what does that result tell the electrician?**

A) PI = 1.5 — a low ratio; suspect moisture or contamination

B) PI = 1.5 — any ratio above 1.0 shows the winding is dry

C) PI = 40 MΩ — the rise in resistance over the ten minutes

D) PI = 0.67 — the one-minute reading divided by the ten-minute

Q8. **What is the purpose of a distribution transformer in an industrial facility?**

A) To step down supply voltage to utilization voltage

B) To filter the harmonic currents produced by the plant's variable frequency drives

C) To correct the plant's overall power factor and reduce the utility demand charge

D) To generate electrical power on site to carry the plant through a utility outage

Q9. **What is a differential pressure (DP) transmitter used for?**

A) To measure absolute pressure in a vessel

B) To control the differential pressure in a boiler

C) To measure flow or level by sensing a pressure difference

D) To measure the voltage difference between two phases

Q10. **What is slip in an induction motor?**

A) The mechanical efficiency of the motor

B) The difference between rated and measured voltage

C) The difference between synchronous and rotor speed

D) The starting current relative to full-load current

Q11. **An RTD (Pt100) temperature sensor reads -40°C in an oven that is clearly at room temperature (~22°C). A technician measures approximately 84 Ω at the RTD terminals at the DCS. What is the MOST likely fault?**

A) The DCS analog card has failed

B) RTD element has failed open

C) RTD calibration drift from overheating

D) RTD extension wires are partially shorted

Q12. **What personal protective equipment is required for energized work on 600 V equipment?**

A) No PPE is needed at 600 V, which is low voltage

B) Leather work gloves and a hard hat are enough

C) Safety glasses and CSA-approved work boots only

D) Arc-rated PPE with insulated gloves and tools

Q13. **What is the formula for three-phase power?**

A) P = V × I (same as single phase)

B) P = √3 × VL × IL × PF (line values)

C) P = VL × IL / √3

D) P = 3 × V × I × PF (where V = line voltage)

Q14. **What happens if a current transformer (CT) secondary is open-circuited while the primary is energized?**

A) The primary current collapses toward zero because the transformer's magnetic circuit is broken

B) Dangerously high voltage appears on the secondary

C) The loss of secondary current is detected and the protective relay trips the feeder offline

D) Nothing hazardous happens — the CT simply stops producing an output until it is reconnected

Q15. **What is the difference between a "fail open" (FO) and "fail closed" (FC) control valve?**

A) FO opens at full stroke; FC closes at full stroke

B) FO is for liquid service; FC is for gas service

C) FO uses a butterfly valve; FC uses a globe valve

D) FO opens on loss of air/power; FC closes on loss

Q16. **What is a timer instruction in PLC ladder logic?**

A) An instruction that restarts the processor after a set time

B) An instruction that changes a done bit at a preset time

C) A network module that synchronizes the clocks of several PLCs

D) A hardware clock module that sequences machine operations

Q17. **What is a control valve Cv (flow coefficient)?**

A) The valve stroke in percentage per volt of signal

B) The valve's closing velocity in m/s

C) The valve's flow capacity at 1 psi pressure drop

D) The calibration verification number of the valve

Q18. **What is the synchronous speed of a 4-pole, 60 Hz induction motor?**

A) 1200 RPM

B) 1800 RPM

C) 900 RPM

D) 3600 RPM

Q19. **What is arc flash and what CSA standard governs arc flash safety?**

A) Explosive energy release from an arc fault; governed by CSA Z462

B) Static discharge from rotating machinery; governed by CSA Z267

C) Electrical shock from AC systems; governed by CEC C22.1

D) Voltage surge in power lines; governed by IEEE 1584

Q20. **What is program scan time and why does it matter in safety-critical applications?**

A) The time between scheduled PLC maintenance — longer scan = more wear

B) The time for the PLC to boot after power loss

C) The time for one complete scan cycle — limits response speed

D) The time for the PLC to communicate with the HMI

Q21. **Under CEC Section 18, which wiring method is acceptable in a Class I, Zone 1 (formerly Division 1) hazardous location?**

A) Threaded rigid metal conduit with flameproof fittings

B) Rigid PVC conduit with ordinary solvent-weld couplings

C) Electrical metallic tubing with set-screw couplings

D) NMD90 non-metallic sheathed cable stapled to framing

Q22. **A wound-rotor induction motor is connected with external resistance in the rotor circuit. After the resistance is short-circuited, motor RPM increases from 1,680 to 1,740 under the same load. What does this confirm?**

A) The motor had a shorted stator winding

B) Synchronous speed of this motor is 1,740 RPM

C) External resistance was increasing slip and reducing speed

D) The motor was single-phasing before the resistance change

Q23. **What is a bus bar in electrical distribution?**

A) A grounding conductor in a substation

B) A solid conductor forming a common connection point

C) A type of fuse used in high-current applications

D) The main communication network in a PLC system

Q24. **What is a delta-wye (Δ-Y) transformer connection and what is its advantage?**

A) Wye primary reduces short circuit current; Delta secondary increases voltage

B) The wye secondary provides a grounded neutral

C) Both delta and wye are identical in performance

D) Delta is for HV primary; Wye is only for motor loads

Q25. **What is an HMI in automation?**

A) Hydraulic Motor Interlock — a safety system

B) Human-Machine Interface — operator display and controls

C) High Motor Interface — motor speed feedback device

D) High-speed Memory Interface — PLC memory module

Q26. **In a 4-wire wye system, why must the neutral conductor and the bonding conductor be kept separate everywhere downstream of the service?**

A) Joining them would raise the line-to-neutral voltage at single-phase loads

B) The bonding conductor is always smaller than the neutral it would parallel

C) The neutral carries normal load current, which would then flow on enclosures

D) The neutral must be kept isolated from earth at every point in the system

Q27. **What does FLA or FLC stand for on a motor nameplate?**

A) Full Line Ampacity

B) Frequency Limit Adjustment

C) Full Load Amperes

D) Fuse Load Allowance

Q28. **What is impedance (Z) in AC circuits?**

A) The maximum instantaneous current an AC circuit can carry when it is driven at resonance

B) The phase angle in degrees by which the current waveform lags or leads the applied voltage

C) The purely resistive component of the circuit, which an ohmmeter reads directly across the load

D) The total opposition to current flow in an AC circuit

Q29. **What is a ground fault circuit interrupter (GFCI) and where is it required?**

A) A device that protects branch circuit conductors from sustained overload and thermal damage

B) A device that limits the arc-flash incident energy released at the panel during a fault

C) A fuse element that clears bolted short circuits and high-current faults only

D) A device that opens the circuit on a ground-fault imbalance of about 6 mA

Q30. **A technician uses a clamp meter to measure current on a 3-phase motor with star (Y) connection. Each phase measures 18A. What is the line current feeding this motor?**

A) 54A (18 × 3)

B) 18A (line = phase)

C) 31.2A (18 × √3)

D) 10.4A (18 / √3)

Q31. **What routine maintenance does a wound-rotor induction motor need that a squirrel-cage motor does not?**

A) Undercutting the mica between the segments of the commutator

B) Cleaning and adjusting the centrifugal starting switch contacts

C) Checking the tightness of the cast rotor bars in the end rings

D) Inspecting and replacing the brushes riding on the slip rings

Q32. **Two jobs are planned on the same 600V system: one beside an exposed fixed circuit part inside a switchboard, the other beneath an exposed movable overhead conductor. How do the CSA Z462 limited approach boundaries for the two jobs compare?**

A) The movable conductor job has the larger limited approach boundary

B) The fixed circuit part job has the larger limited approach boundary

C) Both jobs share one limited approach boundary set by voltage alone

D) Limited approach boundaries begin only above 750 volts to ground

Q33. **What is the purpose of a safety relay or safety PLC?**

A) To supply backup control power so the machine keeps running when the main PLC loses supply

B) To protect the PLC input and output cards from voltage spikes and inductive switching transients

C) To automatically reset and restart the machine as soon as the emergency stop button is pulled out

D) To monitor safety functions and force a safe state on fault

Q34. **Under CEC Section 28, how is motor branch-circuit overcurrent protection sized relative to the motor full-load current (FLC)?**

A) At exactly 100% of the motor's FLC

B) At 80% of the motor service factor current

C) Above FLC, using the Code multipliers

D) Only Class J current-limiting fuses are permitted

Q35. **What is a 4–20 mA current loop signal used for?**

A) A DC power supply circuit that feeds the PLC I/O modules

B) A control signal sent from a VFD to the motor stator windings

C) A standard analog signal for transmitting process variables

D) A digital protocol for networking PLCs on a plant control bus

Q36. **What is a soft starter?**

A) A magnetic starter with a slow-close contactor

B) An electronic device that ramps up voltage during starting

C) A variable autotransformer for motor starting

D) A motor with low-friction bearings for smooth starting

Q37. **What is an equipotential bonding mat used for in industrial work?**

A) To protect workers from step and touch potential

B) To bond the conduit and raceway system back to the driven ground electrode at the service entrance

C) To measure the resistance of bonding conductors and the ground grid during commissioning tests

D) To provide a clean insulated surface for storing rubber gloves and live-line tools between tasks

Q38. **What is the purpose of a dynamic braking resistor connected to a VFD?**

A) To start the motor faster by providing additional energy

B) To protect the VFD from voltage spikes

C) To dissipate regenerated energy as heat during deceleration

D) To improve power factor during motor starting

Q39. **An overhead hoist keeps stalling its general-purpose motor as it takes up a full load. The replacement motor is specified as NEMA Design D. What does that designation change?**

A) Higher full-load efficiency, with slip lower than Design B

B) A larger frame for the same horsepower, and the same torque

C) High locked-rotor torque, with high slip at rated load

D) Lower starting current, with torque unchanged from Design B

Q40. **What is a 2-wire vs 4-wire transmitter?**

A) 2-wire transmitters are for digital signals; 4-wire for analog

B) 2-wire is loop-powered; 4-wire has a separate power supply

C) Wire count refers to the number of calibration points only

D) 2-wire = only positive terminal; 4-wire = both terminals

Q41. **What is the purpose of a power factor correction capacitor bank?**

A) To regulate voltage at the service entrance

B) To filter voltage harmonics from VFDs

C) To supply reactive power locally

D) To store energy for emergency power

Q42. **Switchgear feeding a large pump motor has been replaced. A phase rotation tester at the motor starter reads the same A-B-C sequence as before the work. Why must the motor still be bumped and the shaft direction observed before the pump goes back into service?**

A) Because the tester reads sequence only above the motor's rated voltage

B) Because the supply sequence alone does not fix which way the shaft turns

C) Because bumping is the only way to confirm the overload is sized right

D) Because a rotation tester cannot read the sequence on a loaded circuit

Q43. **A 442A electrician measures a single-phase 120V AC circuit drawing 15A with a power factor of 0.75 lagging. What is the true (real) power consumed?**

A) 1,800 W

B) 1,190 W

C) 2,400 W

D) 1,350 W

Q44. **What is the purpose of a star-delta (Y-Δ) starter?**

A) To step up voltage for high-voltage motor starting

B) To reduce starting current by starting in star, then switching to delta

C) To allow the motor to run in both clockwise and counterclockwise directions

D) To vary motor speed by changing the number of poles

Q45. **What is the role of a motor nameplate?**

A) To specify the bearing lubrication intervals and the preventive maintenance schedule

B) To provide rated data for installation and protection

C) To display the manufacturer's brand and catalogue number for identification purposes only

D) To record the warranty period and the serial number needed when filing a claim

Q46. **A 4-20mA loop-powered transmitter reads 3.6mA with the process at 0% (minimum range). What does this indicate and what should the technician do FIRST?**

A) Transmitter fault — verify loop supply and zero trim

B) Broken signal wire — replace immediately

C) Normal — some transmitters operate below 4mA at process zero

D) PLC analog input card has failed

Q47. **What is the difference between AC and DC electricity?**

A) AC reverses direction periodically; DC flows one way

B) AC is only for power transmission; DC is for motors

C) DC is safer than AC for all applications

D) AC always has higher voltage than DC

Q48. **What is a normally open (NO) contact in PLC ladder logic?**

A) A contact used only in emergency stop circuits

B) A contact that passes logic when its reference bit is TRUE

C) A contact that passes current only when its coil is de-energized

D) A physical contact that is open by default in the field

Q49. **What does CEC Section 2 address?**

A) Wiring methods and cable types

B) Special installations such as pools

C) General rules and administration

D) Grounding and bonding of systems

Q50. **What is the HART protocol and what is its primary advantage?**

A) Hardwired Addressable Relay Technology — a relay-based hardwired interlock scheme

B) High-speed Analog Real-Time Transmission — faster than a standard 4-20 mA loop

C) A fibre-optic link that replaces the 4-20 mA loop between field and control room

D) Highway Addressable Remote Transducer — digital data on the same 4-20 mA loop

Answer Key & Explanations

Q1. Answer: D

A force overrides the actual state of an I/O point inside the processor. A forced input is read by the program as whatever value was set, no matter what the field device is doing; a forced output is driven to the forced state, no matter what the ladder logic solved. That is genuinely useful for proving wiring with the process shut down, and dangerous the moment the plant is live: a forced output can energize a starter or stroke a valve with no rung calling for it, and a forced input can hold an interlock or a limit switch in a state the field device never sent. Do not assume that switching the processor out of run and back has cleared them - list the forces in the programming software, log them, and remove them before the machine is handed over. Forcing does nothing to the output module's load rating, it does not hold the processor in a mode, and it does not store the program to memory.

Q2. Answer: C

Power factor is the ratio of real power to apparent power: PF = P/S = cos θ. Running the ratio the other way, apparent power over real power, is not power factor - that quotient is always 1 or greater and can never be a power factor. Ratios built on reactive power give something else again: reactive power over apparent power is the reactive factor, sin θ, and reactive power over real power is the tangent of the phase angle. Low power factor, typically caused by inductive loads such as motors, means the current is out of phase with the voltage, so more current is drawn for the same real power - raising I^2R conductor losses and utility demand charges.

Q3. Answer: A

A small voltage unbalance shows up as a much larger current unbalance. Current unbalance runs roughly 6 to 10 times the voltage unbalance, on a percent basis. These three readings average 27.7A and the worst phase is 3.3A off that average, so the current unbalance is about 12% - which points back to a voltage unbalance of only about 1 to 2% at the supply, small enough to be missed unless the line voltages are actually measured. Worn bearings cause noise and drag but do not unbalance the phase currents. Single-phasing would show one phase near zero. A genuine overload raises all three phases together, and this motor is running unloaded.

Q4. Answer: A

Current transformers (CTs) step down large primary currents (e.g., 1000A) to a standardized 5A or 1A secondary for safe measurement by metering, energy meters, and protective relays.

Q5. Answer: A

A voltage tester that has failed reads zero on a live conductor exactly as it reads zero on a dead one. A blown fuse, a broken lead, a flat battery, a damaged range switch - none of them announce themselves, and every one of them turns the instrument into a device that says 'safe' about everything. Proving the tester on a known live source immediately before the absence-of-voltage test, taking the test, then proving it again on the known live source afterwards is what establishes that the zero reading meant the conductors were dead rather than that the instrument was dead. The second proving matters as much as the first, because an instrument can fail during the test. This live-dead-live check is a step in establishing an electrically safe work condition, after the disconnecting devices have been opened, locked out and visually verified. Selecting the right range matters but detects nothing about a failed instrument, nothing needs discharging from the leads, and a second person watching does not verify the meter.

Q6. Answer: D

CEC Rule 8-102 requires that voltage drop not exceed 3% in a feeder or branch circuit, and not exceed 5% overall from the supply side of the consumer's service to the point of utilization. This is mandatory 'shall not exceed' language, not a design suggestion, and an inspector can write it up. At 6.7% the branch circuit is over the limit, so the conductor is undersized for a 50 m run at 20 A. Voltage drop is current times resistance, and the only listed action that reduces the resistance of the run is a larger conductor: more cross-sectional area means less resistance per metre and less drop. A regulator at the load end hides the symptom while leaving the same heating in the conductor, and changing the load does not correct an installation that fails the rule.

Q7. Answer: A

The polarization index is the ten-minute insulation resistance divided by the one-minute reading: 120 ÷ 80 = 1.5. It is a ratio, not a difference, and the later reading goes on top - inverting the two gives 0.67, which is the usual arithmetic slip on this test. On a clean, dry winding the resistance keeps climbing through the ten minutes as the absorption current decays, and the ratio comes out well clear of the minimum for that insulation. A ratio of 1.5 means the reading barely moved, which is the signature of moisture or of surface contamination carrying leakage current across the winding; it does not confirm a dry winding, because a ratio just above unity is exactly what a wet or dirty winding gives. Clean and dry the winding and repeat the test, and record the winding temperature - insulation resistance is strongly temperature-dependent, so readings must be corrected to a common temperature before they are compared or trended.

Q8. Answer: A

Distribution transformers step down the high-voltage utility supply (e.g., 13.8 kV) to the utilization voltages used by plant equipment. Common industrial: 13.8 kV → 600V for large motors/feeders, 600V → 120/208V for lighting and controls.

Q9. Answer: C

DP transmitters measure the difference between two pressure taps. With an orifice plate restriction, DP indicates flow (square root relationship). For level measurement, DP corresponds to liquid column height × density.

Q10. Answer: C

Slip is expressed as a percentage: Slip = (Ns - Nr) / Ns × 100%. At no load, slip is small. At full load, typical slip is 2–5%. Slip allows the rotor conductors to experience a changing magnetic field, inducing the rotor current that creates torque.

Q11. Answer: D

A shorted RTD lead appears as lower resistance than actual temperature. A short circuit somewhere in the 3-wire RTD extension leads reduces measured resistance below actual, making the DCS calculate a lower (incorrect) temperature. Pt100 at 22°C ≈ 108.6 Ω, but at −40°C it is 84.27 Ω (IEC 60751) — the ~84 Ω measured means a partial short in the leads is bypassing ~25 Ω of the element's true resistance, so the DCS computes −40°C.

Q12. Answer: D

Energized work on a 600 V industrial system can produce a severe arc flash. Under CSA Z462 the protection is chosen one of two ways, and the two must not be mixed: either an incident-energy analysis is done and the arc rating of the clothing and equipment must equal or exceed the incident energy calculated for that task, or the arc-flash PPE tables in Z462 are used and the clothing must meet the minimum arc rating those tables specify for that task on that equipment. Either route ends in the same kit - arc-rated clothing with a face shield or an arc-rated hood, rubber insulating gloves with leather protectors rated for the system voltage, and insulated tools. Energized work also requires a documented justification for not de-energizing and an energized-work permit.

Q13. Answer: B

Three-phase real power: $P = \sqrt{3} \times V_L \times I_L \times \cos\theta$. Where V_L = line-to-line voltage, I_L = line current, $\cos\theta$ = power factor. This applies to balanced 3-phase systems.

Q14. Answer: B

With the secondary open, the primary current acts solely as magnetizing current, fully magnetizing the CT core. The core saturates and the collapsing magnetic flux induces extremely high, potentially lethal voltage spikes (thousands of volts) on the open secondary terminals.

Q15. Answer: D

Fail-safe position is determined by process safety analysis. Fail open (FO): the spring pushes the valve open on loss of instrument air (e.g., cooling water valves must stay open on failure). Fail closed (FC): the spring closes the valve (e.g., fuel gas valves must close on failure).

Q16. Answer: B

Timer instructions count elapsed time against a preset value, and each type drives its done bit differently. A TON, timer on-delay, sets its done bit once the accumulated time reaches the preset while the rung stays true. A TOF, timer off-delay, sets its done bit as soon as the rung goes true and resets it once the accumulated time reaches the preset after the rung goes false - nothing is set at the preset. An RTO, retentive timer, keeps its accumulated time when the rung goes false and must be cleared with a RES instruction. Timers are used for timed sequences, delay functions and time-based control.

Q17. Answer: C

Cv is the flow capacity of a valve — the volume of water (US gallons per minute) that flows through it with a 1 psi pressure drop. Sizing: $C_v = Q \times \sqrt{SG/\Delta P}$ where Q is flow in gpm, SG is specific gravity, ΔP is pressure drop in psi. Larger Cv = larger flow capacity.

Q18. Answer: B

Synchronous speed = $(120 \times f) / P = (120 \times 60) / 4 = 1800$ RPM. A 4-pole motor runs at approximately 1750 RPM (slip accounts for the difference).

Q19. **Answer: A**

Arc flash is an explosive electrical discharge that releases intense heat (up to 20,000°C), pressure, and UV/IR radiation. CSA Z462 (Workplace Electrical Safety) establishes arc flash hazard analysis and PPE requirements.

Q20. **Answer: C**

Scan time — the duration of one complete read-execute-write cycle — determines how quickly the PLC and its control loop respond to process changes. In safety applications, if the scan time is too long, a hazardous condition may persist for longer than acceptable. Safety PLCs have deterministic, certified scan times.

Q21. **Answer: A**

A Zone 1 location can hold an explosive gas atmosphere during normal operation, so the wiring must neither become a source of ignition nor let an ignition inside the system propagate out to the surrounding atmosphere. Threaded rigid metal conduit with flameproof (explosion-proof) fittings does both: the long threaded joints act as flame paths that cool escaping gases below the ignition temperature of the atmosphere outside. The other Zone 1 method used every day in Canadian plants is hazardous-location certified cable, such as HL-marked Teck 90 terminated in certified cable glands. Set-screw tubing, solvent-welded PVC and non-metallic sheathed cable offer no flame path and no certified gland, so none of them is acceptable.

Q22. **Answer: C**

External rotor resistance increases slip, reducing speed. Wound-rotor motors use external resistance to limit starting current and control speed. When resistance is removed (short-circuited) for normal operation, slip decreases and speed approaches synchronous speed. 1,800 RPM synchronous at 60 Hz for a 4-pole motor; 1,740 RPM represents normal running slip.

Q23. **Answer: B**

Bus bars are solid flat or rectangular copper or aluminum conductors that carry large currents in switchgear, switchboards, and MCCs. Multiple circuits tap off the bus bar, which serves as their common connection point and is sized for the full load current.

Q24. **Answer: B**

A delta-wye transformer has the primary wound in delta, with no neutral, and the secondary wound in wye with a neutral point that is grounded. That gives a 4-wire secondary carrying both the three-phase line-to-line voltage (for example 600 V) and a line-to-neutral voltage (347 V, since $600/\sqrt{3} = 346.4$), so single-phase loads and the system ground reference are taken from the same transformer. Wrong answers: a wye primary does not reduce short-circuit current and a delta secondary does not raise voltage — available fault current is set by the supply impedance and by the transformer's impedance and turns ratio, not by the winding configuration; a wye secondary is not restricted to motor loads, since its whole value is that it also supplies line-to-neutral single-phase loads; and the two connections are not identical in performance, because only the wye secondary offers a neutral. For contrast, a wye-delta bank places the neutral on the primary side only and leaves a three-wire secondary with no neutral, while a delta-delta bank has no neutral on either side. The delta winding also gives triplen harmonic currents a path to circulate, keeping them out of the other winding. Red Seal point: 600 V/347 V systems in Canada use delta-wye transformers so that 347 V single-phase lighting branch circuits can be supplied from the same transformer as 600 V three-phase motor loads.

Q25. **Answer: B**

An HMI (Human-Machine Interface) allows operators to monitor and control automated processes via touchscreens or panel displays. It shows process status, alarms, and trends, and allows setpoint entry.

Q26. **Answer: C**

The neutral of a wye system is a current-carrying circuit conductor: it returns the unbalanced portion of the single-phase load current, and it does so continuously in normal operation. The bonding conductor carries current only during a fault. The two are joined at exactly one place - at the supply, where the system is grounded. Join them again anywhere downstream and the neutral current divides between the neutral conductor and every parallel metal path back to that point: conduit, raceways, equipment enclosures, structural steel, piping. Those parts are then carrying load current all day, which is what makes it objectionable rather than merely untidy, and it also corrupts ground-fault detection, since a residual device can no longer tell returning load current from fault current. The neutral is not isolated from earth - it is deliberately grounded, but at one point only. Line-to-neutral voltage is set by the transformer, not by where the bond is made.

Q27. **Answer: C**

FLA (Full Load Amperes) or FLC (Full Load Current) is the motor's rated current drawn at nameplate voltage and full mechanical load. Used for sizing conductors, overload protection, and motor starter components.

Q28. **Answer: D**

Impedance $Z = \sqrt{R^2 + (XL-Xc)^2}$. It is the complete opposition to AC current — the vector sum of resistance (R), inductive reactance (XL), and capacitive reactance (Xc) at a specific frequency.

Q29. **Answer: D**

A GFCI compares the current leaving on the ungrounded conductor with the current returning on the neutral; any difference is going to ground, possibly through a person. A Class A device must interrupt the circuit when that imbalance is 6 mA or more, and must not interrupt it when the imbalance is 4 mA or less. That is why a GFCI protects people while a fuse or breaker, sized to protect the conductor, will never open at milliamp levels. The threshold and the trip time are two separate specifications: the maximum permitted trip time is an inverse curve, several seconds at the 6 mA threshold and only about 20 ms once the fault current passes a few hundred milliamps. In Ontario, Rule 26-704 requires Class A GFCI protection for 5-15R and 5-20R receptacles installed outdoors within 2.5 m of finished grade, including block-heater receptacles in parking lots.

Q30. **Answer: B**

In a star (Y) connected motor, line current equals phase current. In Y connection: $I_{line} = I_{phase} = 18A$. Only voltage differs: $V_{line} = V_{phase} \times \sqrt{3}$. This is opposite to delta: in delta, $I_{line} = I_{phase} \times \sqrt{3}$, but $V_{line} = V_{phase}$. The clamp meter on each supply line will read 18A.

Q31. **Answer: D**

A wound rotor carries a three-phase winding whose ends are brought out to slip rings on the shaft. Carbon brushes ride on those rings to connect the external resistance during starting and to short it out for running. Brushes and ring surfaces are the only routinely wearing electrical contact in the machine: brushes shorten and must be replaced before the pigtail bottoms out, spring pressure has to stay in range, and the rings themselves are inspected for grooving, glazing and uneven wear and dressed when needed. A squirrel-cage rotor has no winding brought out, no rings and no brushes, so none of this applies to it. The other three belong to other machines. A commutator, with mica to be undercut between its segments, is a DC machine part - slip rings are continuous and are never undercut. A centrifugal starting switch belongs to a split-phase or capacitor-start single-phase motor. Cast bars in end rings are the squirrel-cage construction itself and are not field-serviceable.

Q32. **Answer: A**

CSA Z462 tabulates approach boundaries by system voltage band, and inside each band it lists two different limited approach distances: a larger one for exposed movable conductors and a smaller one for exposed fixed circuit parts. A movable conductor can swing, sag or be dragged toward a worker, so the standard keeps people further back from it. Voltage alone therefore does not settle the boundary — the worker must also establish whether the exposed live part is fixed or movable. The limited approach boundary is the line an unqualified person may not cross unless advised of the hazard and escorted by a qualified person, and it applies at 600V just as it does at higher voltages.

Q33. **Answer: D**

Safety relays/PLCs (designed and rated to IEC 62061 or ISO 13849-1) monitor safety functions such as E-stops, light curtains, and guard interlocks using redundant, self-diagnostic circuits. On fault detection, they cut power to hazardous functions so the machine reaches a safe state before harm occurs. **In Canada those standards reach the shop floor through CSA Z432, Safeguarding of Machinery**, which requires safety-related parts of control systems on newly manufactured machinery to provide functional safety performance as determined by ISO 13849-1 or IEC 62061.

Q34. **Answer: C**

Motor branch-circuit overcurrent protection is covered by CEC Section 28 (Rule 28-200), not Section 26. Because motor starting current can reach 6-8 times FLC, the branch-circuit device is sized above FLC using the Code multipliers so it can ride through the starting inrush. The separate running overload protection - selected at not more than 125% of FLC where the marked service factor is 1.15 or greater, and not more than 115% otherwise, per Rule 28-306 - is what protects the motor from sustained overcurrent.

Q35. **Answer: C**

4–20 mA current loop signals are the standard analog method of transmitting process variables such as temperature, pressure, flow and level in industrial systems. 4 mA represents 0% of the measured range and 20 mA represents 100%, and because the live zero sits at 4 mA a reading of 0 mA is unambiguously a broken wire or a dead transmitter rather than a legitimate low reading. Because the signal is a current rather than a voltage, it is unaffected by conductor voltage drop and is far less susceptible to induced noise over long cable runs.

Q36. Answer: B

A soft starter uses SCRs (thyristors) to gradually ramp up voltage to the motor during starting, limiting inrush current and mechanical stress (belt, coupling, gearbox) compared to DOL starting.

Q37. Answer: A

Equipotential bonding mats ensure that the floor, equipment, and person are at the same electrical potential, preventing dangerous current flow through the worker during a ground fault. They are commonly used on metal grating and in substations, where step and touch potential hazards are greatest.

Q38. Answer: C

When a VFD decelerates a motor, the motor regenerates energy back to the VFD's DC bus. If that energy cannot be returned to the grid and the bus voltage rises above the limit, the dynamic braking module switches in a resistor to dissipate the excess energy as heat.

Q39. Answer: C

The design letter describes the shape of the speed-torque curve built into the rotor, not the size or the enclosure of the machine. Design B is the general-purpose rotor - normal locked-rotor torque and low slip at rated load - so its speed barely moves as load comes on. Design D uses a high-resistance rotor bar; NEMA's own material describes brass or a similar alloy chosen for high resistance, giving high starting torque and high slip. The machine therefore develops far more torque at standstill and yields, slowing noticeably, as the load rises, which is why NEMA lists hoists as the Design D application and why the Design B machine was stalling. The option that leaves torque unchanged from Design B fails on that point alone: torque is exactly what the letter changes. The efficiency option fails on slip - Design D's slip is high, not lower than Design B's, and because rotor loss rises with slip the high-slip machine is the less efficient of the two. And the letter says nothing about frame size, which is a separate designation.

Q40. Answer: B

A 2-wire (loop-powered) transmitter draws its operating power from the 4–20 mA loop current. A 4-wire transmitter has a separate power supply (120V AC or 24V DC) and outputs a 4–20 mA signal independently.

Q41. Answer: C

Inductive loads (motors, transformers) draw reactive power (kVAR) from the utility. Capacitor banks supply this reactive power locally, reducing the reactive current drawn through supply cables, improving power factor, and reducing utility demand charges and penalties.

Q42. Answer: B

A sequence test proves what the supply is doing, not what the shaft will do. A phase rotation tester tells you the order in which the three supply phases reach their peaks at the point where you clipped on. Confirming that after switchgear work is worth doing, because a reversed supply sequence would reverse every motor on the bus at once. But it says nothing about which way one particular shaft turns, and that depends on three more things the tester never sees: how the motor leads were landed in the terminal box, how the winding was connected internally if the motor has been rewound, and whether a coupling or gearbox between motor and pump reverses direction. Only watching the shaft during a momentary bump settles it. The stakes are that a centrifugal pump run backwards still moves some liquid and can look plausible on a flow indicator, so the error survives long enough to do damage. Rotation testers work at normal system voltage and do not need the circuit unloaded, and overload sizing comes from the nameplate full-load current, not from a bump.

Q43. Answer: D

Real power = V × I × PF = 120 × 15 × 0.75 = 1,350 W. Apparent power is what the conductors actually carry: V × I = 1,800 VA, and quoting that figure in watts is the most common error here. Dividing by the power factor instead of multiplying gives 2,400 W, a number larger than the apparent power, which is impossible. Using sin θ in place of cos θ returns about 1,190, but that is the reactive power in VAR, not the real power in watts. Only the real power does useful work; the reactive component is stored and returned each cycle by the circuit inductance.

Q44. Answer: B

Y-Δ starting connects windings in star (Y) first, applying 1/√3 of line voltage per winding, reducing starting current and torque to 1/3 of DOL values. After acceleration, the starter switches to delta (Δ) for full power.

Q45. Answer: B

The motor nameplate provides rated electrical, mechanical, and thermal data: voltage, phase, frequency, FLA, HP/kW, RPM, duty cycle, insulation class, service factor, efficiency, enclosure type, and NEMA/IEC frame designation - all critical for proper motor selection, installation, and protection. The nameplate FLA is the figure the overload device is selected from; it is not the figure the overload device is set to.

Q46. Answer: A

4mA is the minimum live-zero standard — 3.6mA indicates a transmitter fault or calibration drift. In a 4-20mA system, 4mA = 0% process. Below 4mA is outside the valid range and indicates transmitter fault, severely low loop supply voltage (< 12V at transmitter), or calibration drift. Check supply voltage first (should be 24VDC minus line drops), then check zero trim and recalibrate or replace the transmitter.

Q47. Answer: A

AC (alternating current) reverses polarity/direction at the supply frequency (60 Hz in North America = 120 reversals/second). DC (direct current) flows in one constant direction only. AC is used for power distribution; DC for electronics, batteries, and VFD internal circuits.

Q48. Answer: B

In ladder logic, a NO contact (XIC instruction in Allen-Bradley) allows rung continuity when the corresponding bit in memory is 1 (TRUE). It represents the state when the associated coil or input is energized.

Q49. Answer: C

CEC Section 2 (General Rules) is the administrative and general-safety backbone of the code. Its rules - permits and inspection, use of approved equipment (Rule 2-024), marking of equipment (Rule 2-100), working space about electrical equipment (Rules 2-308 and 2-310), and protection of persons and property - apply to every other section unless that section specifically modifies them. The wrong answers name other sections: grounding and bonding is Section 10, wiring methods and cable types are Section 12, and pools, tubs and spas are Section 68. Keep one distinction straight: the definitions, including the definition of "approved", are in Section 0 (Object, scope, and definitions) - Section 2 is where the code tells you to USE approved equipment, not where the term is defined. On Red Seal exams, Section 2 questions often test what "approved" means, and in Canada it is a jurisdictional answer, not a brand-name one: equipment must be certified by a certification organization accredited by the Standards Council of Canada, to a recognized Canadian standard, and must carry that body's Canadian mark - marks such as CSA, cULus, ULC, cETLus or cQPSus. A bare US "UL Listed" mark with no small "c" identifier means certification to US standards only and is not approved for installation in Canada, and the European CE mark is a manufacturer's self-declaration that Canadian regulators do not recognize at all. The authority having jurisdiction is the final arbiter: Ontario adopts the CEC, together with the Ontario Amendments, as the Ontario Electrical Safety Code under O. Reg. 164/99, and the Electrical Safety Authority publishes the list of recognized approval marks it will accept.

Q50. Answer: D

HART (Highway Addressable Remote Transducer) superimposes a digital frequency-shift-keyed signal — roughly ±0.5 mA at the 1200 Hz and 2200 Hz Bell 202 tones — on top of the 4–20 mA loop current. Because the modulation averages to zero, the analog measurement keeps controlling the process while a handheld communicator or an asset-management system reads and writes configuration, runs diagnostics and pulls secondary variables over the same two wires. That is the advantage: digital access with no additional wiring and no disturbance to the analog control signal. Wrong answers: HART is not a faster substitute for the analog loop — the digital channel is deliberately slow, which is why it is used for configuration and diagnostics rather than for closed-loop control; it is not a fibre-optic control-room network, and it does not replace the 4–20 mA loop, it rides on it; and the acronym does not stand for a relay interlock scheme.